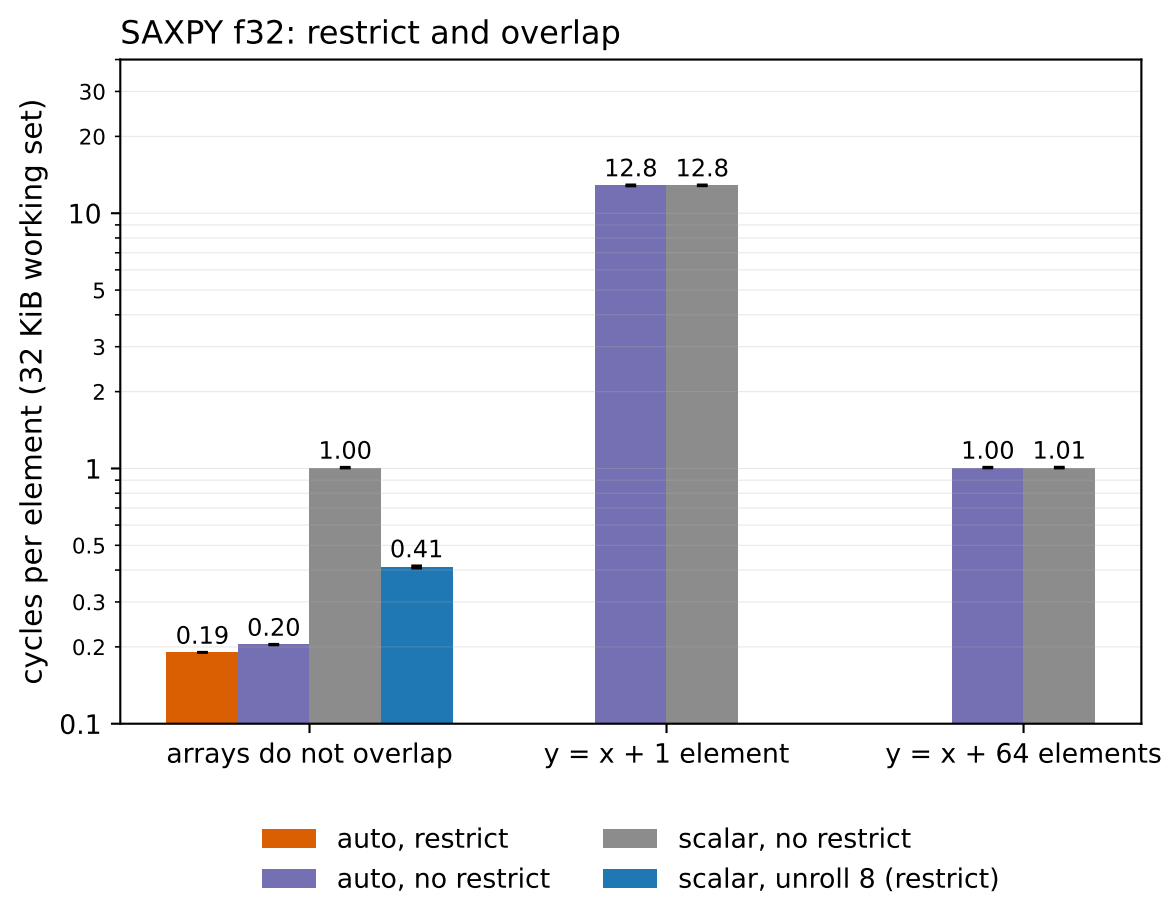
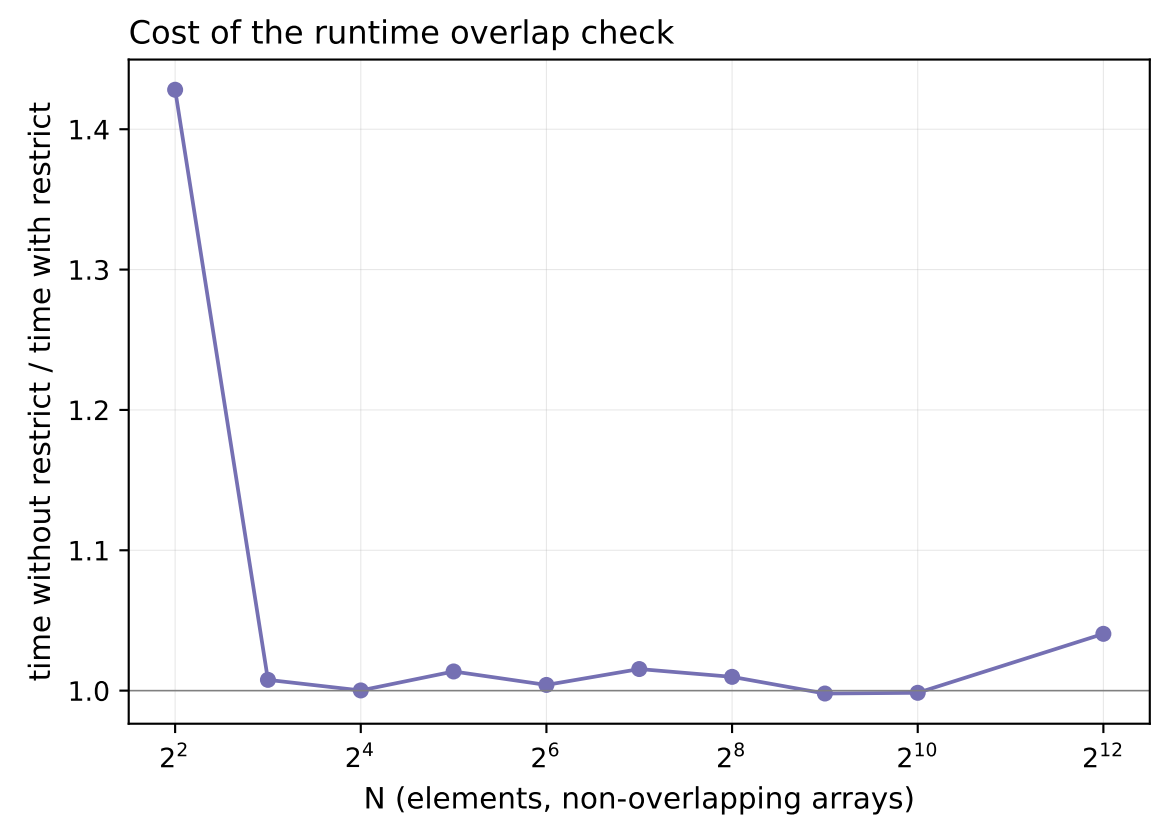

SAXPY f32: restrict and overlap
cycles per element (32 KiB working set)
30
20
10
5
3
2
1
0.5
0.3
0.2
0.1
0.19
0.20
1.00
0.41
12.8
12.8
1.00
1.01
arrays do not overlap
y = x + 1 element
y = x + 64 elements
auto, restrict
auto, no restrict
scalar, no restrict
scalar, unroll 8 (restrict)


Cost of the runtime overlap check
time without restrict / time with restrict
1.4
1.3
1.2
1.1
1.0
2^2
2^4
2^6
2^8
2^{10}
2^{12}
N (elements, non-overlapping arrays)